

Dynamics of Aerial-Balance-Bench

I. DYNAMICS OF THRUST-COMMAND INTERFACE

A. Ball's Absolute Acceleration

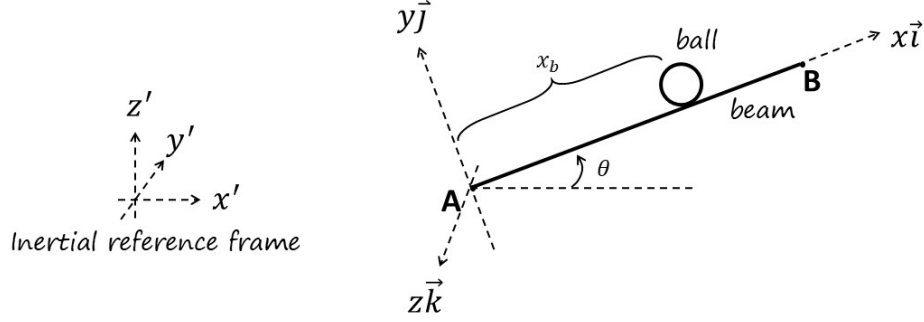


Fig. 1. Illustration of the ball's kinematics.

Let $\vec{a}_b'$ be the absolute acceleration of the ball under the inertial reference frame, then we have

$$\vec{a}_b' = \vec{a}_A' + \dot{\vec{\omega}} \times \vec{x}_b + \vec{\omega} \times (\vec{\omega} \times \vec{x}_b) + 2\vec{\omega} \times \vec{v}_b + \vec{a}_b, \quad (1)$$

where

$$\begin{aligned} \vec{\omega} &= \dot{\theta} \vec{k}, \\ \vec{x}_b &= x_b \vec{i}, \\ \vec{v}_b &= \dot{x}_b \vec{i}, \\ \vec{a}_b &= \ddot{x}_b \vec{i}, \end{aligned} \quad (2)$$

and $\vec{a}_A'$ is the absolute acceleration of the point A. Substituting Equation (2) into Equation (1), we have

$$\begin{aligned} \vec{a}_b' &= \vec{a}_A' + x_b \ddot{\theta} \vec{j} - x_b \dot{\theta}^2 \vec{i} + 2\dot{\theta} \dot{x}_b \vec{j} + \ddot{x}_b \vec{i} \\ &= \vec{a}_A' + (\ddot{x}_b - x_b \dot{\theta}^2) \vec{i} + (x_b \ddot{\theta} + 2\dot{\theta} \dot{x}_b) \vec{j}. \end{aligned} \quad (3)$$

B. Ball's Dynamics Equation

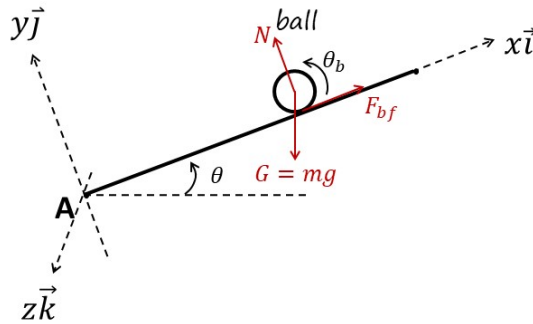


Fig. 2. Illustration of the ball's dynamics.

The ball is rolling on the beam, we have

$$J_b \ddot{\theta}_b = M_b = F_{bf} r_b, \quad (4)$$

where J_b denotes the ball's moment of inertia, θ_b is the ball's rotation angle, r_b is the radius of the ball, and F_{bf} is the friction between the ball and the beam.

Assuming that there is no slipping between the ball and the beam, then we have

$$\dot{\theta}_b = -\frac{\dot{x}_b}{r_b}, \quad \ddot{\theta}_b = -\frac{\ddot{x}_b}{r_b}. \quad (5)$$

Substituting Equation (5) into Equation (4), we have

$$\begin{aligned} J_b \left(-\frac{\ddot{x}_b}{r_b} \right) &= F_{bf} r_b, \\ F_{bf} &= -\frac{J_b}{r_b^2} \ddot{x}_b. \end{aligned} \quad (6)$$

Because the coordinate frame is attached to point A , which is generally non-inertial, the translational acceleration of A must be retained in the force balance. Let

$$\vec{a}_A' = a_{A,i}' \vec{i} + a_{A,j}' \vec{j}. \quad (7)$$

To compute $a_{A,i}'$ and $a_{A,j}'$, introduce the inertial basis $\{\vec{e}_x, \vec{e}_z\}$, where $\vec{e}_x$ is horizontal and $\vec{e}_z$ is vertical upward. Since the beam makes an angle θ with the horizontal, the rotating basis attached to the beam can be written as

$$\vec{i} = \cos \theta \vec{e}_x + \sin \theta \vec{e}_z, \quad \vec{j} = -\sin \theta \vec{e}_x + \cos \theta \vec{e}_z. \quad (8)$$

If the absolute acceleration of point A in the inertial frame is

$$\vec{a}_A' = \ddot{x}_A \vec{e}_x + \ddot{z}_A \vec{e}_z, \quad (9)$$

then its components along the rotating axes are obtained by projection:

$$\begin{aligned} a_{A,i}' &= \vec{a}_A' \cdot \vec{i} = \ddot{x}_A \cos \theta + \ddot{z}_A \sin \theta, \\ a_{A,j}' &= \vec{a}_A' \cdot \vec{j} = -\ddot{x}_A \sin \theta + \ddot{z}_A \cos \theta. \end{aligned} \quad (10)$$

Considering the dynamics along the $x\vec{i}$ axis, we have

$$\begin{aligned} m_b(a_{A,i}' + \ddot{x}_b - x_b \dot{\theta}^2) &= F_{bf} - m_b g \sin \theta, \\ m_b(a_{A,i}' + \ddot{x}_b - x_b \dot{\theta}^2) &= -\frac{J_b}{r_b^2} \ddot{x}_b - m_b g \sin \theta, \\ \left(\frac{J_b}{r_b^2} + m_b \right) \ddot{x}_b &= -m_b g \sin \theta + x_b m_b \dot{\theta}^2 - m_b a_{A,i}'. \end{aligned} \quad (11)$$

Considering the dynamics along the $y\vec{j}$ axis, we have

$$\begin{aligned} N - m_b g \cos \theta &= m_b(a_{A,j}' + x_b \ddot{\theta} + 2\dot{\theta} \dot{x}_b), \\ N &= m_b g \cos \theta + m_b(a_{A,j}' + x_b \ddot{\theta} + 2\dot{\theta} \dot{x}_b). \end{aligned} \quad (12)$$

The acceleration terms in Equations (11) and (12) can be further specialized by using the rigid-beam constraint between the two beam endpoints:

$$\vec{r}_B' = \vec{r}_A' + L\vec{i}. \quad (13)$$

Differentiating Equation (13) twice with respect to time yields

$$\vec{a}_B' = \vec{a}_A' + L\ddot{\theta}\vec{j} - L\dot{\theta}^2\vec{i}, \quad (14)$$

or equivalently,

$$\vec{a}_A' = \vec{a}_B' + L\dot{\theta}^2\vec{i} - L\ddot{\theta}\vec{j}. \quad (15)$$

Therefore, the acceleration components of point A can be written as

$$a_{A,i}' = a_{B,i}' + L\dot{\theta}^2, \quad a_{A,j}' = a_{B,j}' - L\ddot{\theta}. \quad (16)$$

a) Case: if Point B is fixed.: If point B is fixed in the inertial frame, then $\vec{a}_B' = 0$, and Equation (16) becomes

$$a_{A,i}' = L\dot{\theta}^2, \quad a_{A,j}' = -L\ddot{\theta}. \quad (17)$$

Substituting Equation (17) into Equations (11) and (12) gives

$$\left(\frac{J_b}{r_b^2} + m_b \right) \ddot{x}_b = -m_b g \sin \theta + m_b (x_b - L) \dot{\theta}^2. \quad (18)$$

$$N = m_b g \cos \theta + m_b [(x_b - L) \ddot{\theta} + 2\dot{\theta} \dot{x}_b]. \quad (19)$$

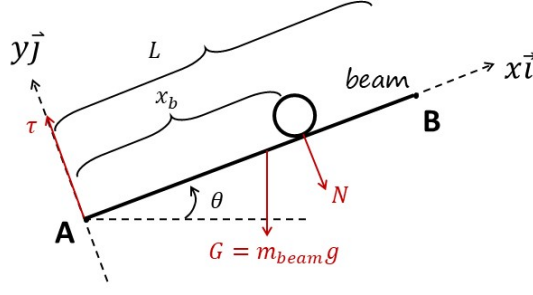


Fig. 3. Illustration of the beam's dynamics.

C. Beam's Dynamics Equation

For the general case in which both endpoints A and B may move, the rotational dynamics of the beam should be written using the angular momentum balance about the accelerating point B :

$$J_{beam}\ddot{\theta} - m_{beam}\frac{L}{2}a'_{B,j} = N(L - x_b) + m_{beam}g \cos \theta \frac{L}{2} - \tau L, \quad (20)$$

where J_{beam} denotes the beam's moment of inertia about point B , L is the length of the beam, m_{beam} is the beam's mass, N represents the pressure from the ball, τ represents the force exerted by the rope perpendicular to the beam, and $a'_{B,j}$ is the component of point B 's absolute acceleration along the $y\vec{j}$ axis.

If point B is fixed, then $a'_{B,j} = 0$, and Equation (20) reduces to

$$J_{beam}\ddot{\theta} = N(L - x_b) + m_{beam}g \cos \theta \frac{L}{2} - \tau L. \quad (21)$$

Substituting Equation (19) into Equation (21), we have

$$\begin{aligned} [J_{beam} + m_b(L - x_b)^2]\ddot{\theta} = & (m_bg \cos \theta + 2m_b\dot{\theta}\dot{x}_b)(L - x_b) + m_{beam}g \cos \theta \frac{L}{2} \\ & - \tau L. \end{aligned} \quad (22)$$

D. Drone's Dynamics Equation

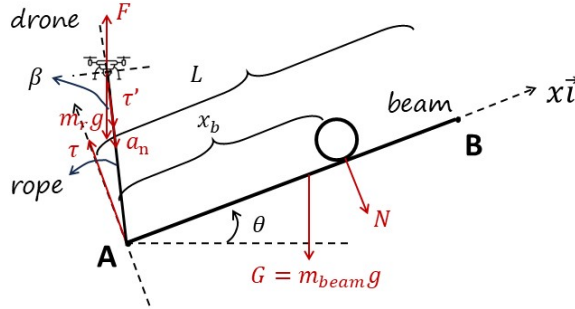


Fig. 4. Illustration of the drone's dynamics.

When the rope makes a nonzero angle β with the vertical direction, the rope tension acting on the drone has both horizontal and vertical components. Therefore, a self-consistent drone model should explicitly account for how the horizontal rope force is balanced. In this work, we model the drone as a system constrained to move only along the vertical direction.

The drone is assumed to move only along a vertical guide, or equivalently, a sufficiently fast low-level controller is assumed to exactly cancel the horizontal rope force and keep the horizontal position of the drone fixed. Let R_x denote the horizontal constraint force acting on the drone. Since the beam is initially horizontal, the fixed horizontal coordinate of the drone is chosen as $x_r = x_B - L$.

Under this assumption, the rope geometry becomes

$$l \sin \beta = L(1 - \cos \theta), \quad z_r - z_B = l \cos \beta - L \sin \theta. \quad (23)$$

Differentiating the vertical relation in Equation (23) twice with respect to time yields

$$\ddot{z}_r - \ddot{z}_B = -l(\cos \beta \dot{\beta}^2 + \sin \beta \ddot{\beta}) + L(\sin \theta \dot{\theta}^2 - \cos \theta \ddot{\theta}). \quad (24)$$

The horizontal and vertical force balances of the constrained drone are

$$0 = R_x + \tau' \sin \beta, \quad m_r \ddot{z}_r = F - \tau' \cos \beta - m_r g. \quad (25)$$

The rope force transmitted to the beam along the $y\vec{j}$ axis is

$$\tau = \tau' \cos(\beta - \theta). \quad (26)$$

Hence,

$$R_x = -\tau' \sin \beta = -\tau \frac{\sin \beta}{\cos(\beta - \theta)}, \quad (27)$$

$$\tau = \frac{\cos(\beta - \theta)}{\cos \beta} (F - m_r g - m_r \ddot{z}_r). \quad (28)$$

Substituting Equation (24) into Equation (28), we obtain

$$\tau = \frac{\cos(\beta - \theta)}{\cos \beta} [F - m_r g - m_r \ddot{z}_B + m_r l(\cos \beta \dot{\beta}^2 + \sin \beta \ddot{\beta}) - m_r L(\sin \theta \dot{\theta}^2 - \cos \theta \ddot{\theta})]. \quad (29)$$

If point B is fixed, then $\dot{z}_B = 0$ and $\ddot{z}_B = 0$. Equation (29) reduces to

$$\tau = \frac{\cos(\beta - \theta)}{\cos \beta} [F - m_r g + m_r l(\cos \beta \dot{\beta}^2 + \sin \beta \ddot{\beta}) - m_r L(\sin \theta \dot{\theta}^2 - \cos \theta \ddot{\theta})]. \quad (30)$$

Since the drone thrust is limited, the dynamic is required to satisfy the constraint $F \leq F_{\max}$.

E. Core Dynamic Model

In this interface, the system input is the drone's vertical thrust F . During execution. This interface offers the most direct actuation and fastest response, but it also yields the strongest coupling between ball balancing and low-level flight control. For short, the core dynamic model can be written as

$$(\frac{J_b}{r_b^2} + m_b) \ddot{p}_b = m_b(p_b - L) \dot{\theta}^2 - m_b g \sin \theta, \quad (31a)$$

$$[J_{beam} + m_b(L - p_b)^2] \ddot{\theta} = (m_b g \cos \theta + 2m_b \dot{\theta} \dot{p}_b)(L - p_b) + \frac{L}{2} m_{beam} g \cos \theta - \tau L, \quad (31b)$$

$$\tau = [F - m_r g + m_r l(\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta}) - m_r L(\sin \theta \dot{\theta}^2 - \cos \theta \ddot{\theta})] \frac{\cos(\beta - \theta)}{\cos \beta}, \quad (31c)$$

$$\sin \beta = \frac{L}{l}(1 - \cos \theta), \quad (31d)$$

$$0 \leq |F| \leq F_{\max}. \quad (31e)$$

Here, the high-level command directly enters the beam dynamics through the rope-induced force τ , so this interface retains the richest coupling with the drone actuation dynamics.

II. DYNAMICS OF VELOCITY-COMMAND INTERFACE

A. Beam's kinematics Equation

In this interface, we assume that $v_h = \dot{z}_B$ denotes the vertical velocity of the Point B and $v_r = \dot{z}_r$ represents the drone vertical velocity, and they are treated as system's inputs. Because the beam is initially horizontal and both Point B and the drone move only along the vertical direction, the same geometric constraints as Equation (23) apply when the rope is taut:

$$l \sin \beta = L(1 - \cos \theta), \quad z_r - z_B = l \cos \beta - L \sin \theta. \quad (32)$$

Differentiating Equation (32) with respect to time yields

$$l \cos \beta \dot{\beta} = L \sin \theta \dot{\theta}, \quad v_r - v_h = -l \sin \beta \dot{\beta} - L \cos \theta \dot{\theta}. \quad (33)$$

Eliminating $\dot{\beta}$ from Equation (33), we obtain

$$L \dot{\theta} = \frac{\cos \beta}{\cos(\beta - \theta)} (v_h - v_r). \quad (34)$$

If Point B is fixed, $v_h = 0$, then we obtain

$$L \dot{\theta} = -v_r \frac{\cos \beta}{\cos(\beta - \theta)}. \quad (35)$$

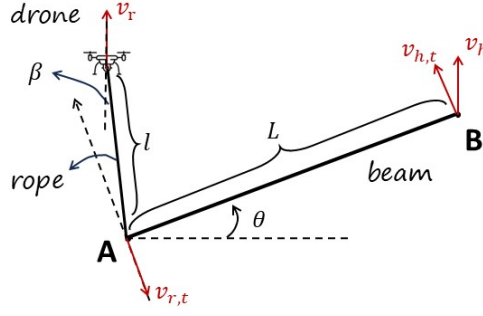


Fig. 5. Illustration of the drone's hierarchical kinematics.

B. Core Dynamic Model

In this interface, the system input is the drone's vertical velocity v_r . The ball's dynamics are same as that of the thrust-command interface, so the core dynamic model can be written as

$$\left(\frac{J_b}{r_b^2} + m_b\right)\ddot{p}_b = m_b(p_b - L)\dot{\theta}^2 - m_b g \sin \theta, \quad (36a)$$

$$L\dot{\theta} = -v_r \frac{\cos \beta}{\cos(\beta - \theta)}, \quad (36b)$$

$$\sin \beta = \frac{L}{l}(1 - \cos \theta), \quad (36c)$$

$$|\dot{v}_r| \leq a_{\max}. \quad (36d)$$

Compared with direct thrust command, this interface removes most of the system-level complexity from the high-level policy and achieves a compromise between control authority and design simplicity. Using this control interface, the controller only needs to generate feasible velocity commands, while the low-level controller handles the corresponding velocity tracking. However, this hierarchical architecture may introduce longer control delays.

III. DYNAMICS OF POSITION-COMMAND INTERFACE

A. Beam's Kinematics Equation

Let d_h and d_r denote the Point B and drone vertical displacements relative to their initial positions:

$$d_h = z_B - z_{B,0}, \quad d_r = z_r - z_{r,0}. \quad (37)$$

Assuming the beam is initially horizontal and the rope is initially vertical, we have $\theta(0) = 0$, $\beta(0) = 0$, and

$$z_{r,0} - z_{B,0} = l. \quad (38)$$

Substituting $z_B = z_{B,0} + d_h$ and $z_r = z_{r,0} + d_r$ into the geometric constraint in Equation (23) and then using Equation (38), we obtain the direct relation between the beam angle and the relative vertical displacement:

$$d_h - d_r = L \sin \theta + l(1 - \cos \beta), \quad \sin \beta = \frac{L}{l}(1 - \cos \theta). \quad (39)$$

Equation (39) shows that the relative displacement $d_h - d_r$ is composed of the vertical motion of the beam endpoint, $L \sin \theta$, and the additional vertical shortening caused by the rope inclination, $l(1 - \cos \beta)$. Therefore, β acts only as an auxiliary geometric variable, while the main input-output relation is between θ and $d_h - d_r$.

If Point B is fixed, then $d_h = 0$, then we have

$$d_r = l(\cos \beta - 1) - L \sin \theta. \quad (40)$$

B. Core Dynamic Model

In this interface, the system input is the drone's vertical displacement d_r . This command is tracked by a low-level position controller. Since the ball's dynamics are same as that of the thrust-command interface and velocity-command interface, the core dynamic model can be written as follows:

$$(\frac{J_b}{r_b^2} + m_b)\ddot{p}_b = m_b(p_b - L)\dot{\theta}^2 - m_b g \sin \theta, \quad (41a)$$

$$d_r = l(\cos \beta - 1) - L \sin \theta, \quad (41b)$$

$$\sin \beta = \frac{L}{l}(1 - \cos \theta), \quad (41c)$$

$$|\ddot{d}_r| \leq a_{\max}. \quad (41d)$$

This interface provides the highest level of abstraction and has the easiest dynamics formulation. However, its effective response depends on the derivatives of the commanded displacement and on the tracking behavior of the low-level position controller, making it more susceptible to lag than the thrust- and velocity-command interfaces.